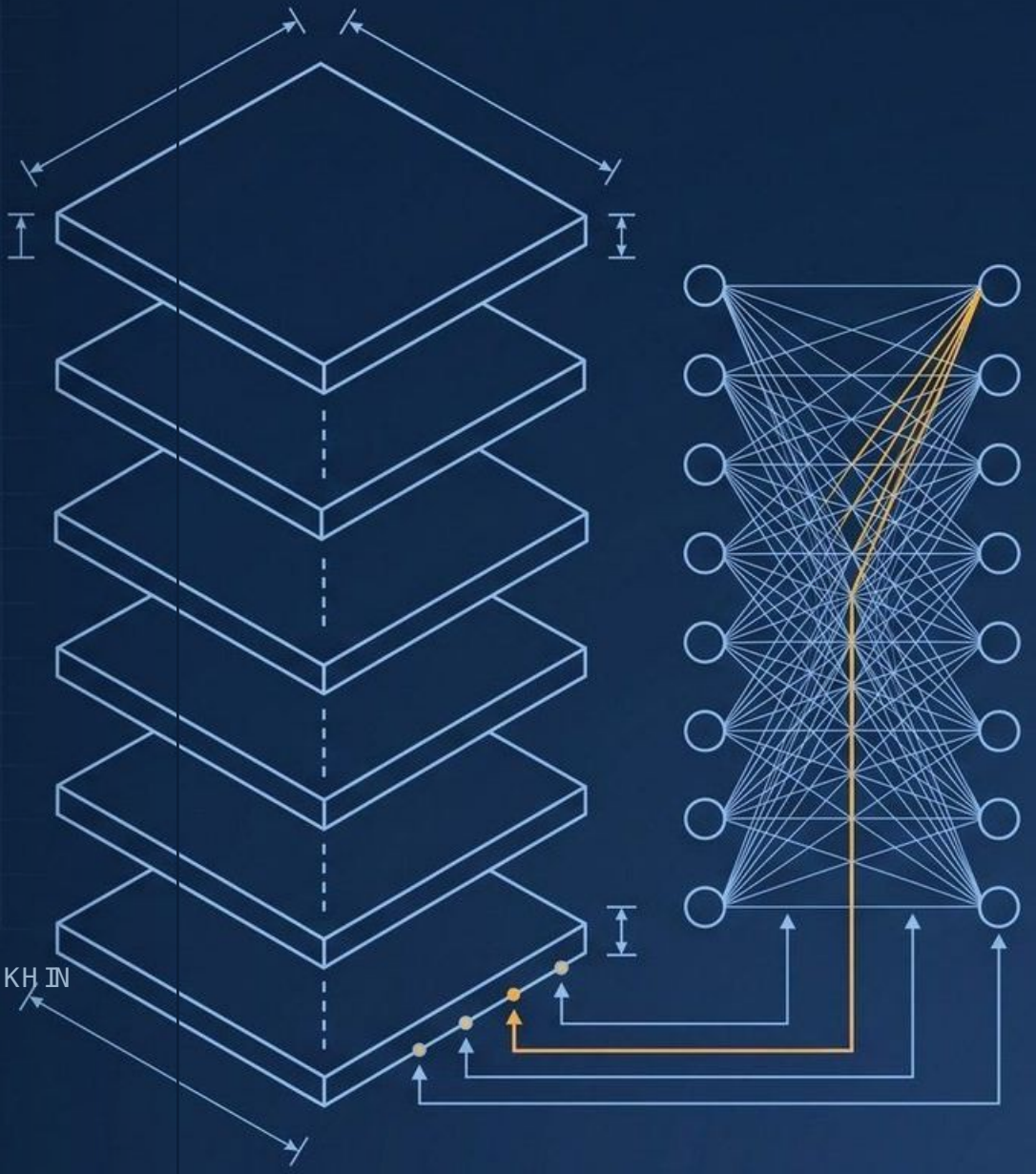
ARXIV 1706.03762 · NEURIPS 2017

Attention Is All You Need

The Transformer, explained:
one paper, no recurrence, no convolutions,
and a training run measured in days

VASWANI · SHAZEER · PARMAR · USZKOREIT · JONES · GOMEZ · KAISER · POLOSUKHIN
GOOGLE BRAIN · GOOGLE RESEARCH



DRAWING INDEX

What this drawing set covers

01 **The bottleneck** — SHEETS 03–04
Why recurrent and convolutional models paid for word order with time

02 **The architecture** — SHEET 05
Encoder, decoder, and the six identical layers stacked on each side

03 **Attention, step by step** — SHEETS 06–10
Scaled dot-product, multi-head, the three places it is used, and positional encoding

04 **Training and results** — SHEETS 11–13
The recipe, the BLEU numbers against the field, and what the ablations taught

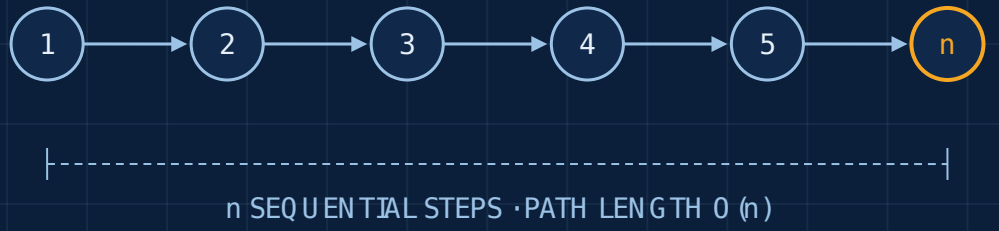
05 **Legacy** — SHEETS 14–15
What was built on top of it, and why the title still holds

§ 01 · THE BOTTLENECK

Before 2017, models paid for word order with time

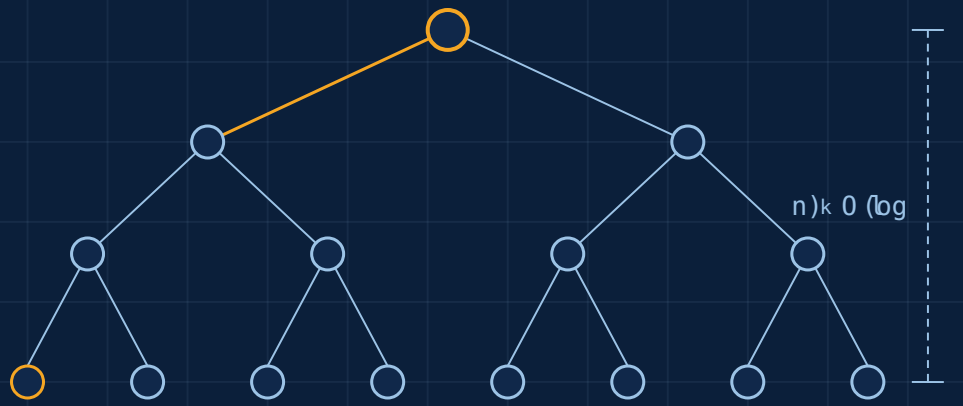
A · RECURRENT (LSTM / GRU)

$h_t = f(h_{t-1}, x_t)$ — every step waits for the previous one



Position 1 reaches position n only through n hops.
Inside one example nothing can run in parallel,
so longer sentences mean longer wall-clock time.

B · CONVOLUTIONAL (ByteNet / ConvS2S)



Parallel inside a layer, but two distant positions
only meet after a stack of kernel-k layers —
the path shrinks to O(log n), it does not vanish.

Both architectures charge a **distance tax** : the farther apart two positions are, the longer the path a signal must travel — and recurrence cannot run in parallel within one training example. The paper's question: what if every path had length 1?

Keep attention, drop everything else

99

> “We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely .”
>
> — ABSTRACT, VASWANI ET AL. 2017

THREE CLAIMS THE PAPER MAKES — AND MEASURES



28.4 BLEU

Better quality

WMT14 EN–DE, over 2 BLEU above the best prior ensembles

O(1)

More parallelizable

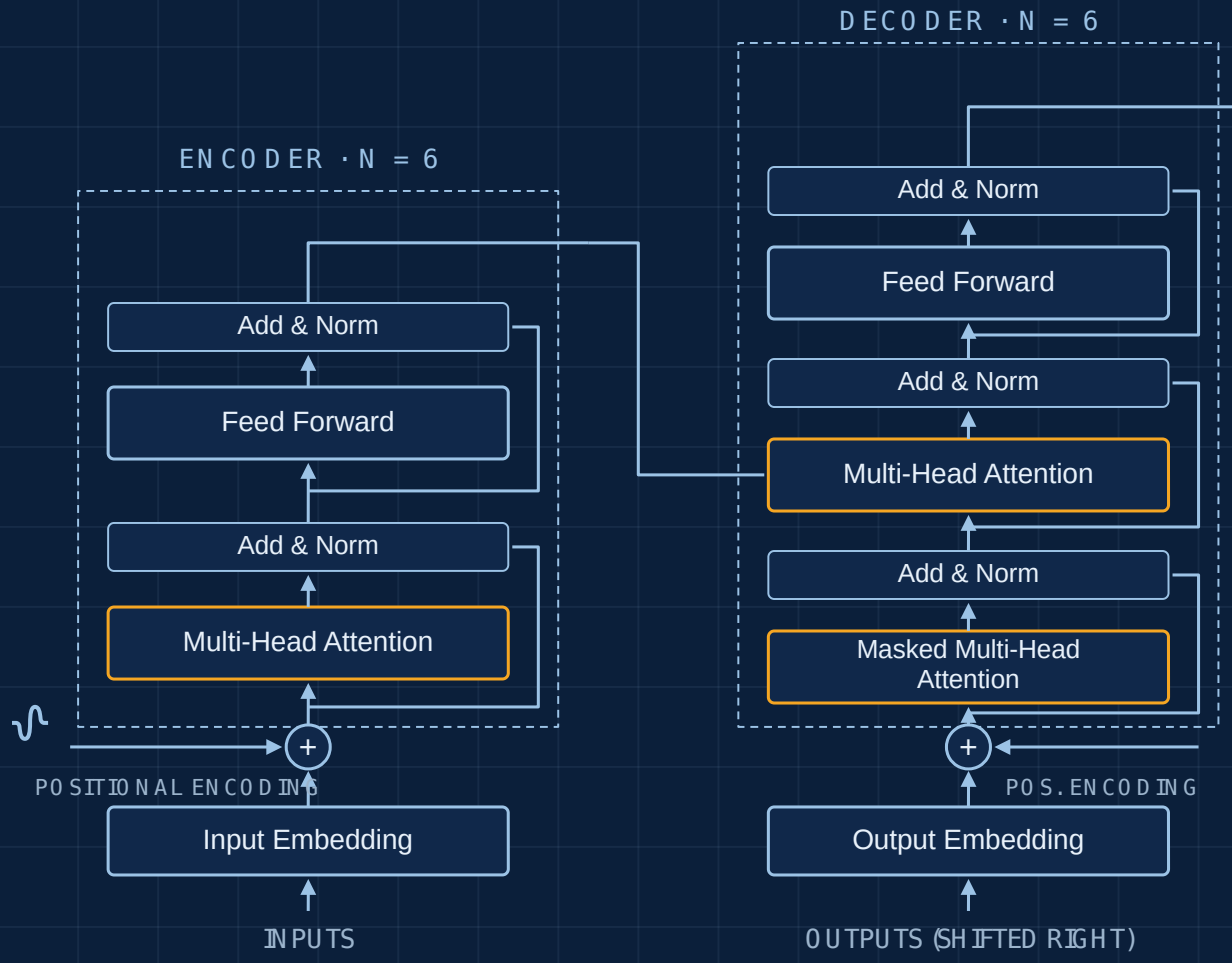
sequential ops per layer — all positions handled at once

3.5 days

Far less training time

8 × P100 GPUs, big model; 41.8 BLEU on WMT14 EN–FR

Six identical layers on each side



- **$d_{model} = 512$**
  One width for every sub-layer output
- **$h = 8 \cdot d_k = d_v = 64$**
  8 heads, each a 64-wide subspace
- **$d_{ff} = 2048$ · position-wise**
  $\mathrm{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$
- **Add & Norm around each sub-layer**
  LayerNorm(x + Sublayer(x))
- **PARAMETERS**
  base 65 M · big 213 M

Scaled dot-product attention

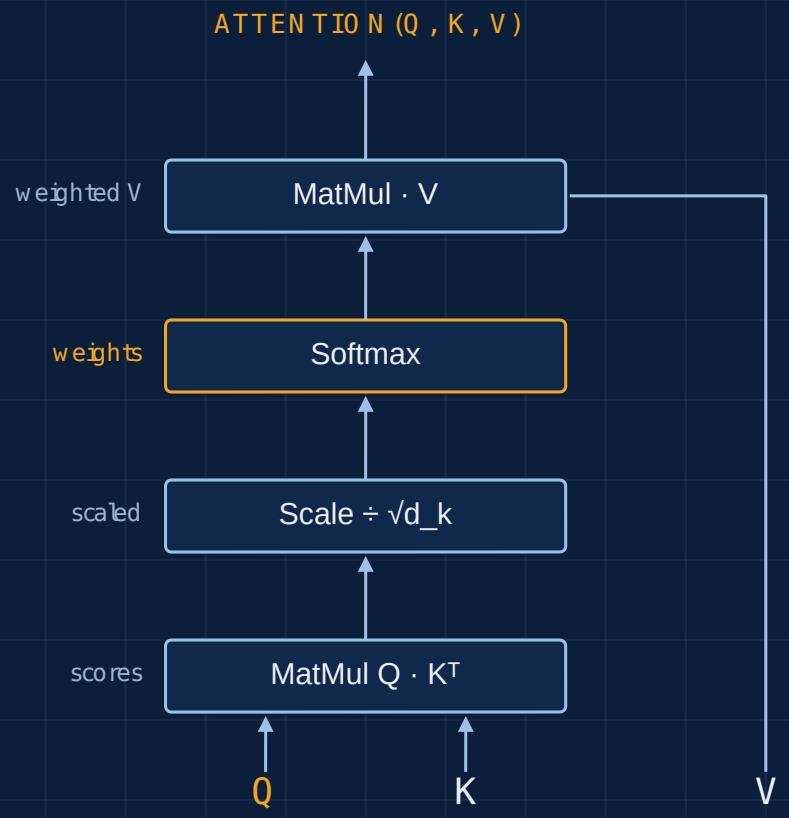
$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})\, V$$

- **Q** — query: what this position is looking for
- **K** — key: what each position offers for matching
- **V** — value: what is passed along once matched

Why divide by √d_k?

With large d_k the dot products grow large and push softmax into regions with tiny gradients; the 1/√d_k factor keeps the scores tame.

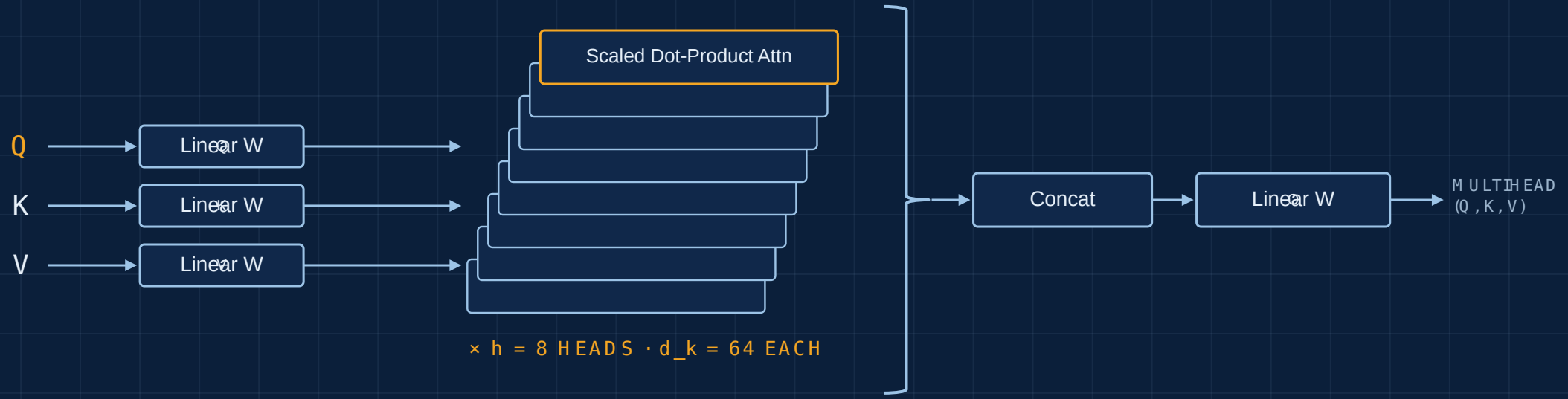
d_k = 64 → divide every score by 8



Multi-head attention: eight views at once

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\, W^O$$

$$\text{head}_i = \text{Attention}(Q\, W_i^Q, K\, W_i^K, V\, W_i^V)$$

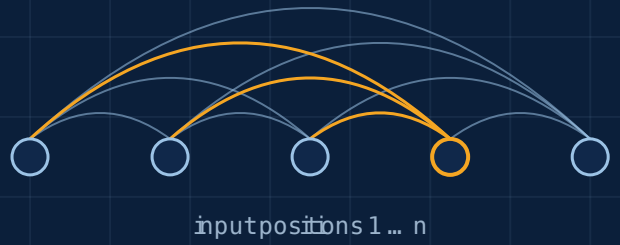


- d_k = d_v = 512 / 8 = 64 — eight reduced heads cost about as much as one full-width head
- Ablation (Table 3): a single attention head is 0.9 BLEU worse than the best setting

The same block is used in three places

A · ENCODER SELF-ATTENTION

Q, K, V ← previous encoder layer

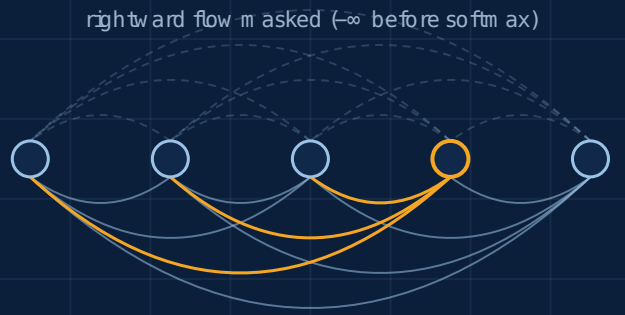


Every position looks at every other position, in both directions, in one step — the O(1) path.

Six of these, stacked, build the encoder's contextual representation.

B · MASKED DECODER SELF-ATTN

Q, K, V ← previous decoder layer

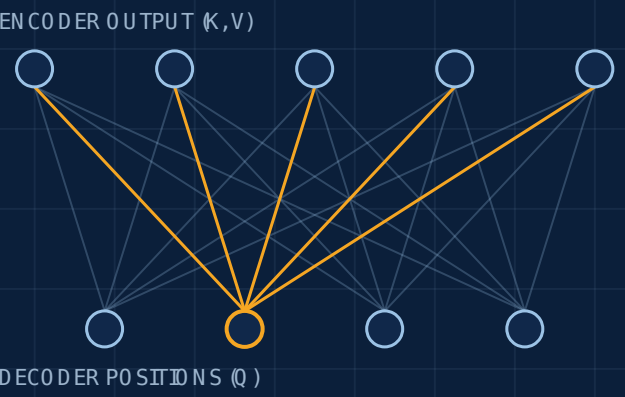


Same block, but position i may only attend to positions ≤ i. The mask keeps generation auto-regressive.

Outputs are shifted right by one position for the same reason.

C · ENCODER–DECODER ATTENTION

Q ← decoder · K, V ← encoder output



Queries come from the decoder; keys and values from the encoder output. Every output position can read the whole input sentence.

This replaces the classic seq2seq encoder–decoder attention.

Injecting order without recurrence

$$PE(pos, 2i) = \sin(pos / 10000^{(2i / d_model)})$$

$$PE(pos, 2i+1) = \cos(pos / 10000^{(2i / d_model)})$$

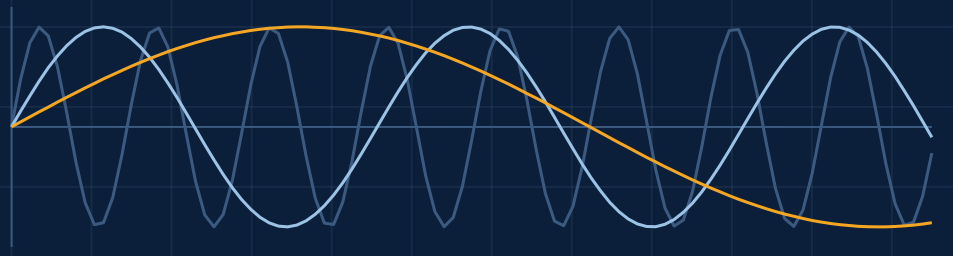
Why sinusoids, not learned positions?

Learned positional embeddings performed nearly identically in the ablation (Table 3).
The sinusoidal version was kept because it may extrapolate to lengths never seen in training.

Input Embedding + Positional Encoding = $d_model = 512$

added, not concatenated — same width, no extra parameters

ONE SINUSOID PER DIMENSION



wavelengths run from 2π to $10000 \cdot 2\pi$

Each dimension is a sinusoid with its own wavelength, so a position gets a unique code. For any fixed offset k, PE(pos+k) is a linear function of PE(pos): relative offsets are easy.

Three costs, four layer types (Table 1)

Layer type	Complexity per layer	Sequential operations	Maximum path length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	**$O(1)$**
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k n)$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n / r)$

n = sequence length · d = representation dimension · k = kernel width · r = neighbourhood size

> **Any two positions are one operation apart**
> Per-layer cost $O(n^2 \cdot d)$ beats recurrence whenever $n < d$ — the usual case for word-piece or BPE sentences. For very long inputs the restricted variant keeps a window r; the path grows to $O(n/r)$, still shorter than $O(n)$.

The spec sheet: data, hardware, schedule

DATA · EN–DE

4.5 M sentence pairs · shared BPE vocabulary ≈ 37 k

DATA · EN–FR

36 M sentences · 32 k word-piece vocabulary

BATCHING

≈ 25 k source + 25 k target tokens per batch

HARDWARE

One machine · 8 × NVIDIA P100 GPUs

SCHEDULE · BASE MODEL

100 k steps · 0.4 s per step · ≈ 12 hours

SCHEDULE · BIG MODEL

300 k steps · 1.0 s per step · 3.5 days

REV 1 · 2017-06-12 · ARXIV 1706.03762 · WMT 2014

OPTIMIZER

Adam · $\beta_1 = 0.9$ · $\beta_2 = 0.98$ · $\varepsilon = 10^{-9}$

LEARNING RATE · warmup_steps = 4 000

$$lrate = d_model^{-0.5} \cdot \min(step^{-0.5}, step \cdot warmup_steps^{-1.5})$$

WARMUP

Linear ramp for 4 000 steps, then decay $\propto 1/\sqrt{step}$

REGULARIZATION

Residual dropout 0.1 (0.3 for the big EN–FR run)

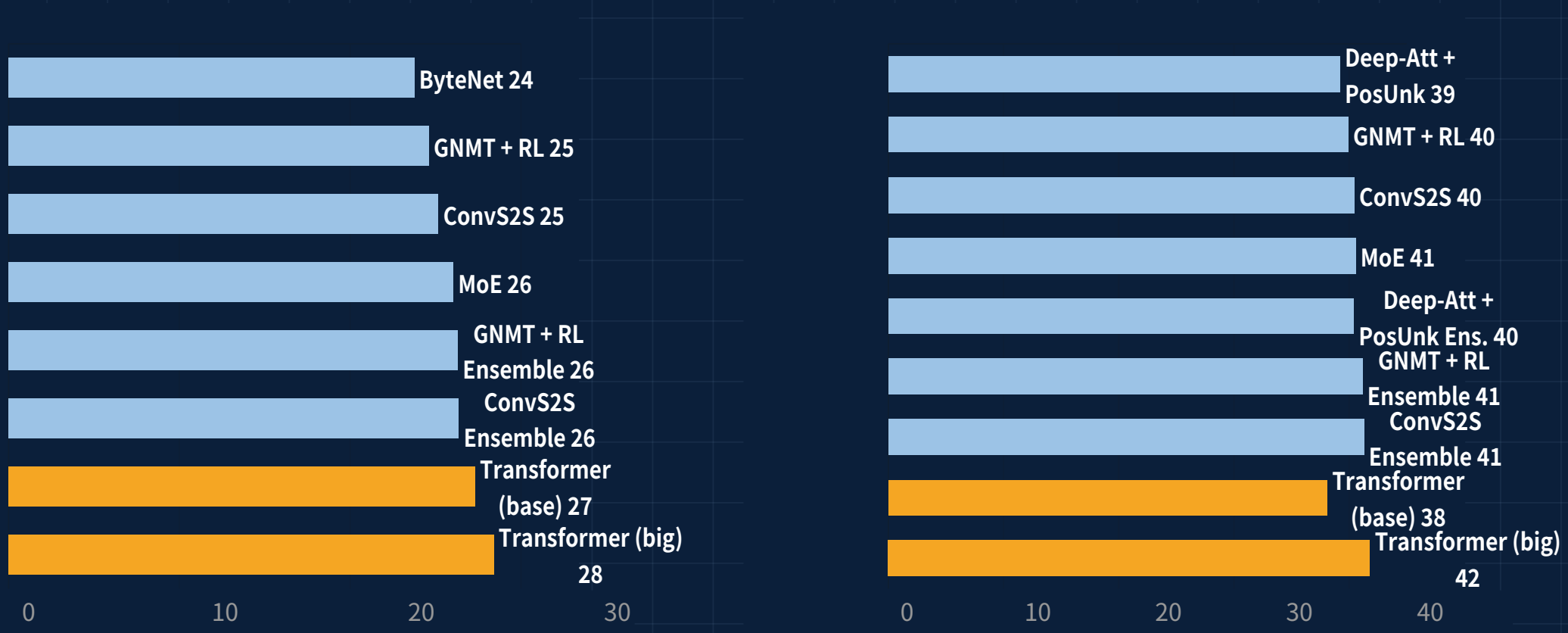
Label smoothing $\varepsilon = 0.1$

MODEL SIZE

Base 65 M parameters · Big 213 M parameters

ALL VALUES FROM §5 OF THE PAPER · 8 × P100 · 3.5 DAYS

State of the art on both WMT 2014 tasks



Training cost: Transformer base used $3.3 \cdot 10^{18}$ FLOPs against $1.8 \cdot 10^{20}$ for the GNMT + RL ensemble — about **55× less compute** for a better EN→DE score.

Source: Vaswani et al. 2017, Table 2 (newstest2014). Single models and ensembles listed together, as in the paper.

What the ablations taught (Table 3)

HEADS · h

A single head is 0.9 BLEU worse than the best setting

KEY SIZE · d_k

Smaller d_k hurts — compatibility is not trivial to compute

MODEL SIZE

Bigger is better; the big model wins on both tasks

REGULARIZATION

Dropout is very helpful in avoiding over-fitting

POSITIONS

Learned vs sinusoidal positions: nearly identical

Each row varies one setting of the base model on EN–DE (newstest2013 dev set)

GENERALIZATION · WSJ PARSING

91.3
F1 · WSJ only
4-layer Transformer

92.7
F1 · semi-supervised
4-layer Transformer

A 4-layer Transformer trained on WSJ alone reaches 91.3 F1; with semi-supervised data it reaches 92.7. The abstract's claim holds: the architecture generalizes beyond translation, large data or small.

§ 05 · LEGACY

What was built on it

SYSTEMS BUILT ON THE TRANSFORMER ARCHITECTURE



250,000+

citations as of 2026 — among the top-ten most-cited this century

8 of 8

authors later left Google to join other companies or found startups

“All You Need Is Love”

the Beatles song the title nods to — one joke, still landing

§ 05 · CLOSING

Attention is all you need — the rest was engineering.

01 Drop recurrence: every position attends to every other in O(1).

02 Eight heads, scaled dot-products, sinusoidal positions do the work.

03 28.4 / 41.8 BLEU, 3.5 days, 8 GPUs — quality and speed at once.

Read the paper: · Vaswani et al., NeurIPS 2017

